

1. Which of the following is not a radical expression that is equivalent to $\sqrt{108}$?

- A. $\sqrt{2} \cdot \sqrt{54}$
- B. $\sqrt{3} \cdot \sqrt{36}$
- C. $\sqrt{5} \cdot \sqrt{21}$
- D. $\sqrt{6} \cdot \sqrt{18}$

2. Which of the expressions from #1 can be rewritten in the form of an integer multiplied by the radical? *Explain.*

For questions 3-4, write two equivalent expressions for each radical expression using the product law of radicals. One of your expressions should simplify to an integer multiplied by a radical.

3. $\sqrt[3]{200} =$ _____

$\sqrt[3]{200} =$ _____

4. $\sqrt{72} =$ _____

$\sqrt{72} =$ _____

5. Rewrite $\sqrt[3]{200}$ as an integer multiplied by a radical.

6. Rewrite $\sqrt{72}$ as an integer multiplied by a radical.

7. Identify all expressions that are equivalent to $-4\sqrt[3]{48}$.

- $2\sqrt[3]{6}$
- $-8\sqrt[3]{6}$
- $-8\sqrt[3]{12}$
- $32\sqrt[3]{2}$
- $-4 \cdot \sqrt[3]{8} \cdot \sqrt[3]{6}$
- $-4 \cdot \sqrt[3]{12} \cdot \sqrt[3]{4}$
- $-4 \cdot (24 \cdot 2)^{\frac{1}{3}}$

For questions 8-10, write an equivalent expression in the form of a constant multiplied by a radical.

8. $\sqrt{392}$

9. $\sqrt[3]{135}$

10. $-3\sqrt[3]{625}$